

Nikhilesh Vijay Chirutala

Full-Stack Software Engineer

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SUMMARY

Full-Stack Software Engineer with 2.8+ years of experience building and maintaining production applications across AI, CRM, trading, and cloud contact center platforms. Strong experience with Python, FastAPI, Django, Flask, JavaScript, React.js, Next.js, REST APIs, MongoDB, AWS, and cloud application development. Experienced in backend and frontend development, API design, business logic, third-party integrations, authentication, RBAC/ACL authorization, cloud automation, and production troubleshooting. Hands-on experience implementing RAG-based chatbot solutions and integrating LLM capabilities for contextual information retrieval. Proven ability to develop scalable backend services, build REST APIs, integrate databases, troubleshoot production issues, test implementations, and support reliable deployments.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Backend & Frameworks: FastAPI, Django, Flask, Node.js, Express.js, REST APIs, API Development, Backend Automation

Frontend: React.js, Next.js, Angular, Tailwind CSS, Bootstrap, Jinja2, Microservices

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Databases: MongoDB, PostgreSQL, Neon DB, SQL

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux

APIs & Integrations: Twilio, GitHub API, REST APIs, Postman, API Integration

Security & Authentication: Authentication, Authorization, RBAC, ACL, JWT

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub, Debugging

WORK EXPERIENCE

Software Engineer, Python • CA One Tech Cloud Inc.

Apr 2025 – Present | Bengaluru, India

- Migrated a high-volume call-recording ETL workflow from a legacy Perl cron architecture to a long-running Python service deployed across 40 production clusters using systemd, feature flags, staged rollout, validation gates, rollback planning, and production monitoring.
- Built the Python integration and control layer between the recording platform and an independently deployed Call Intelligence microservice, implementing service-to-service APIs, eligibility rules, account and campaign controls, daily quotas, configurable retries, response validation, and analysis-result persistence.
- Designed resilient internal API integrations using reusable HTTP clients, connection reuse, configurable retry policies, response validation, and failure handling for reliable communication between distributed Python services.
- Replaced process-local quota enforcement with cluster-wide guarded atomic MySQL upserts, preventing distributed service instances from independently consuming the full daily allowance and ensuring consistent quota enforcement across production clusters.
- Designed multi-cluster database routing and connection pooling for a long-running Python service, routing workloads to the correct MySQL endpoints while reusing connections and isolating failures between cluster configurations.
- Built AWS S3 audit and recovery tooling for recording storage using deterministic object keys and idempotent re-upload workflows, enabling safe recovery of incomplete storage migrations and validation before legacy storage decommissioning.
- Built production observability for Python services using Datadog, StatsD, dashboards, and Sentry error tracking to monitor service health, investigate API failures, diagnose distributed-processing issues, and resolve production incidents.

Software Engineer • Mantra Technologies

Feb 2024 – Mar 2026 | Hyderabad, India

- Worked across four production applications, delivering backend, frontend, API integration, and application features across the stack.
- Built backend modules, REST APIs, and business logic using Python, FastAPI, and MongoDB for application-specific workflows.
- Developed frontend interfaces using React.js, Next.js, Angular, JavaScript, and TypeScript, integrating application workflows with APIs.
- Implemented authentication, RBAC, and ACL-based authorization to enforce access controls across application resources.
- Integrated third-party services including Twilio and GitHub API and supported application deployments across AWS and Vercel.
- Maintained GitHub Actions CI/CD workflows and handled testing, debugging, bug fixes, and production support.

Key Projects:

AI Bull — AI-Powered Trading Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment analysis platform for stock and mutual fund analysis, screening, and back-testing.
- Implemented a Retrieval-Augmented Generation (RAG) based chatbot that retrieves relevant financial and application-specific information to generate contextual responses.

- Integrated the RAG chatbot into application workflows, enabling conversational access to relevant financial information.
- Built FastAPI/Python backend APIs and business logic for financial data processing and application workflows.
- Developed back-testing engines and configurable screening and filtering capabilities for stocks and mutual funds.

CRM.io — AI-Driven CRM System

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, REST APIs

- Developed CRM modules for Contacts, Tasks, communication workflows, and telephony operations using FastAPI and Python.
- Implemented RBAC and ACL authorization to control access to CRM resources based on user permissions.
- Integrated Twilio for telephony and communication functionality and developed server-rendered interfaces using Jinja2.
- Tested APIs and workflows with Postman and resolved bugs across CRM functionality and production workflows.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

Guider — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

Contact Management App — Contacts Storage Application

React.js, JavaScript, Netlify

- Developed a React-based application for creating, viewing, updating, and managing contact information.
- Built reusable UI components and deployed the application on Netlify.

Enterprise REST API Platform — Django and Flask Backend Application

Python, Django, Flask, PostgreSQL, REST APIs

- Built modular REST APIs using Django, Django REST Framework, and Flask for authentication, business logic, and data-driven workflows.
- Integrated PostgreSQL with CRUD operations, database models, relationships, validation, and API error handling.
- Implemented JWT authentication and role-based access control, with API testing and validation using Postman.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur